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1 Optimal transport in Single Cell Biology

1.1 Why optimal transport in Biology

Optimal Transport (OT) has become a powerful framework for single-cell biology because it allows us to align, compare, and integrate cellular populations across diverse experimental conditions and data types.

Aligning cells Single-cell experiments often measure only one modality at a time and cannot track the same cell across time. OT helps overcome these limitations by aligning cellular populations across:

- data measurement modalities
- time points
- tissue slices

This alignment is typically achieved through a learned transport map, which pairs cells across datasets without requiring one-to-one correspondences.

Comparing distributions OT distances provide a principled way to compare entire cell populations, useful for studying differences across:

- species
- conditions

Because biological data has meaningful geometric structure, OT captures differences in distributions more faithfully than simple metric-free comparisons. Comparing distributions across:
Typically uses OT distance. Biological data has meaningful metric structure

Defining a consensus/reference map OT enables the construction of reference atlases by integrating cells from:

- samples from different individuals
- different slices of the same tissue

1.2 Why OT Is Needed: The Multimodality Challenge

Modern sequencing technologies can capture many aspects of the genome (gene expression, chromatin accessibility, DNA methylation, etc.), but:

- multimodal measurements per single cell are very limited
- the same cell cannot be measured at multiple time points
- different modalities measure different features, often in different cells
- cross-modal relationships are complex and context-dependent

As a result, unsupervised or weakly supervised OT methods are used to align modalities and reconstruct multimodal single-cell views.

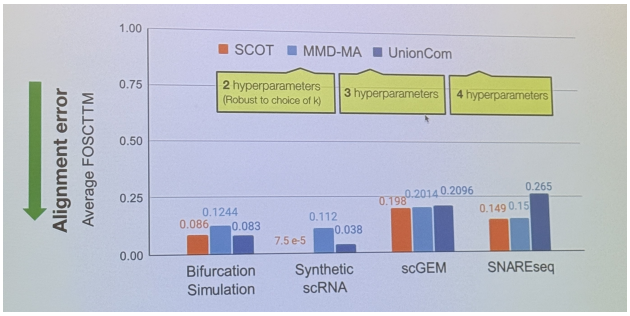


Figure 1: Competitive alignments when hyperparameter tuned

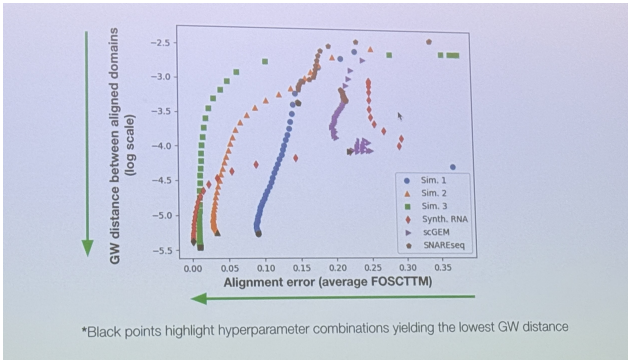


Figure 2: Heuristic for hyperparameter tuning

1.3 Gromov–Wasserstein OT for Cross-Modal Integration

When features differ across modalities, Gromov–Wasserstein OT is particularly valuable because it aligns datasets based on their intrinsic geometric structure, not shared features. To do this, we compute intra-modality cell–cell distances using:

- lower dimensional representation
- k-nearest neighbor graphs
- pairwise shortest paths (Dijkstra's)

$$\min_{\Gamma \geq 0} \sum_{ijkl} (D^X_{ik} - D^Y_{jl})^2 \Gamma_{ij} \Gamma_{kl} - \epsilon H(\Gamma) + \rho KLD(\Gamma_1 | p) + \rho KLD(\Gamma_2 | q)$$

Addressing multi-marginality by picking an anchor domain based on "local autocorrelation", $\text{corr}(z_i, \bar{z}_i)$ where

$$\bar{z}_i = \frac{1}{|N(i)|} \sum_{j \in N(i)} z_j$$

.

Once an OT coupling between datasets is computed, we can exploit it in two ways:

- co-embedding via t-SNE conditioned on the coupling
- Barycentric projection averaging matches, weighted by coupling probabilities

So far we treated GW as fully unsupervised. However, in practice we often have *partial* supervision.

Limited priors on cell–cell relationships Sometimes cells are FACS-sorted using surface markers, which provides coarse cell-type annotations. This is not always possible:

- suitable markers may not exist for all cell types,
- some cell types do not tolerate the sorting protocol.

Limited priors on feature–feature relationships. We also have biological expectations on features:

- highly expressed genes tend to have open chromatin in their regulatory regions,
- specific chemical modifications are known to open or compact DNA.

These pieces of information are sparse and noisy, but they suggest a natural question: can we incorporate such partial supervision into Gromov–Wasserstein OT to guide the alignment? One alternative to GW is Co-optimal transport (COOT).

1.4 Co-optimal Transport (COOT): An Alternative to G-W

We do not have full labels, but we often have noisy, partial biological hints about which cells or features should match, and these can be used as partial supervision.

Recall limited priors on cell-level relationships: Sometimes cells are sorted based on surface markers → gives coarse cell-type annotations – not always possible:

- Don't always have suitable markers to sort by
- Some cell types don't tolerate the experimental protocol

Limited priors on feature relationships:

- We expect the DNA region of a gene with high expression levels to be open for protein binding
- Certain chemical modifications are known to open or close the DNA

Can we introduce partial supervision into GW?
COOT models both levels at once:

- a sample-level coupling Γ^s that aligns cells between the two modalities,
- a feature-level coupling Γ^f that aligns features (e.g. genes or chromatin regions).

The objective aggregates a cost $C(X_{i,a}, Y_{k,b})$ whenever cell i is matched to cell k and feature a is matched to feature b, weighted by $\Gamma^s_{ik}\Gamma^f_{ab}$. Optimizing over both couplings enforces consistency between sample- and feature-level alignments. In practice, one often alternates: fix Γ^f and solve for Γ^s, then fix Γ^s and solve for Γ^f, iterating until convergence.

Cost based on 1D differences

Let $X \in \mathbb{R}^{n\times d}$ and $Y \in \mathbb{R}^{m\times d'}$ denote the two data matrices; w, w' are probability weights over samples and v, v' over features. COOT solves

$$\min_{\Gamma^s \in \Pi(w,w'),\ \Gamma^f \in \Pi(v,v')} \sum_{i,k}\sum_{a,b} C\big(X_{i,a}, Y_{k,b}\big)\, \Gamma^s_{i,k}\, \Gamma^f_{a,b},$$

where $\Pi(w, w')$ and $\Pi(v, v')$ are the sets of couplings with marginals w, w' and v, v', respectively.

Alternatively solve for two coupling matrices, one between samples & one between features.

Redko et al. (2020). CO-Optimal Transport. 34th Conference on Neural Information Processing Systems (NeurIPS).

1.4.1 Unbalanced COOT

Unbalanced COOT relaxes the strict marginal constraints in COOT and instead penalizes deviations with KL terms. This allows aligning datasets with different total mass or missing/extra cells or features.

We now solve for nonnegative couplings $\Gamma^s, \Gamma^f \geq 0$:

$$\min_{\Gamma^s \geq 0,\, \Gamma^f \geq 0} \sum_{i,k} \sum_{a,b} C\big(X_{i,a}, Y_{k,b}\big)\, \Gamma^s_{i,k}\, \Gamma^f_{a,b} + \lambda_1\, \mathrm{KL}\Big((\Gamma^s \mathbf{1}_m) \otimes (\Gamma^f \mathbf{1}_p) \,\Big\|\, w \otimes v\Big) + \lambda_2\, \mathrm{KL}\Big((\Gamma^{s\top} \mathbf{1}_n) \otimes (\Gamma^{f\top} \mathbf{1}_d) \,\Big\|\, w' \otimes v'\Big),$$

where $\mathbf{1}_m$ denotes an m-dimensional all-ones vector, and $\otimes$ is the (outer) tensor product of marginal vectors.

Tran et al. (2023). Unbalanced CO-Optimal Transport. 37th AAAI Conference on Artificial Intelligence (AAAI).

1.5 Augmented Gromov Wasserstein

The idea of Augmented Gromov–Wasserstein (AGW) is to combine two sources of information:

1. a **GW term** that compares pairwise distances between cells inside each dataset, capturing their geometric structure;
2. a **COOT term** that compares individual feature values across datasets via couplings over cells and over features.

1.5.1 (Unbalanced) Augmented Gromov Wasserstein: interpolating (U)COOT and (U)GW formulation

$$\min_{\Gamma^s \geq 0,\, \Gamma^f \geq 0} \underbrace{\alpha \sum_{i,j,k,l} \big(D^X_{ij} - D^{X'}_{kl}\big)^2\, \Gamma^s_{ik}\, \Gamma^s_{jl}}_{\text{(U)GW}} + \underbrace{(1-\alpha) \sum_{i,k} \sum_{a,b} C\big(X_{i,a}, Y_{k,b}\big)\, \Gamma^s_{i,k}\, \Gamma^f_{a,b}}_{\text{(U)COOT}} + \Omega_H(\varepsilon_1, \varepsilon_2) + \Omega_{\mathrm{KLD}}(\rho_1, \rho_2).$$

Demetci et al. (2024) Breaking isometric ties and introducing priors in Gromov–Wasserstein distances. AISTATS.
Baker et al. (2025) SCOT+: A Comprehensive Software Suite for Single-cell Alignment Using Optimal Transport. bioRxiv.

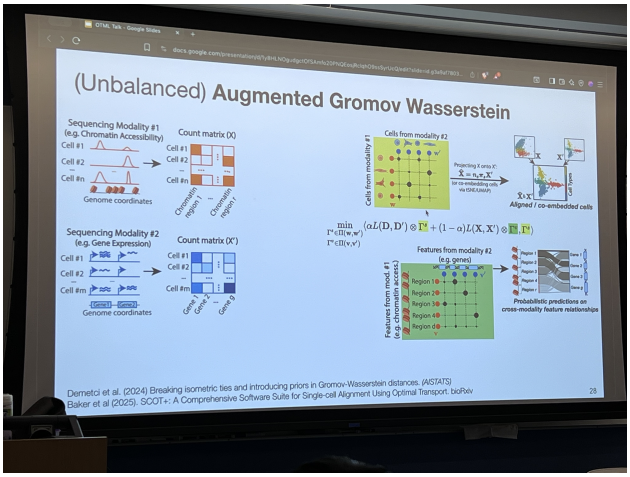


Figure 3: Workflow of AGW

Below is an algorithm for AGW.

Augmented Gromov-Wasserstein (AGW) Algorithm:
Initialize: marginal distributions (default: uniform) over cells and features:

$$w,\ w' \quad \text{over cells,} \qquad v,\ v' \quad \text{over features.}$$

Until convergence:

Optimize cell alignments

$$L_t^s \leftarrow \alpha \sum_{i,j,k,l} \left\| D_{ik}^X - D_{jl}^{X'} \right\|^2 + (1-\alpha) \sum_{i,j} \sum_{a,b} \left\| X_{i,a} - X'_{j,b} \right\|^2 \left(\Gamma_{t-1}^v\right)_{a,b},$$
$$L_t^s \leftarrow M^s \odot L_t^s,$$
$$\Gamma_t^s \leftarrow \text{Sinkhorn}\left(w, w', L_t^s, \varepsilon_1\right).$$

Optimize feature alignments

$$L_t^v \leftarrow \sum_{i,j} \left\| X_{i,a} - X'_{j,b} \right\|^2 \left(\Gamma_t^s\right)_{i,j},$$
$$L_t^v \leftarrow M^v \odot L_t^v,$$
$$\Gamma_t^v \leftarrow \text{Sinkhorn}\left(v, v', L_t^v, \varepsilon_2\right).$$

Demetci et al. (2024) Breaking isometric ties and introducing priors in Gromov–Wasserstein distances. AISTATS.
Baker et al. (2025) SCOT+: A Comprehensive Software Suite for Single-cell Alignment Using Optimal Transport. bioRxiv.

AGW has indeed demonstrated superior performance in multiple settings.

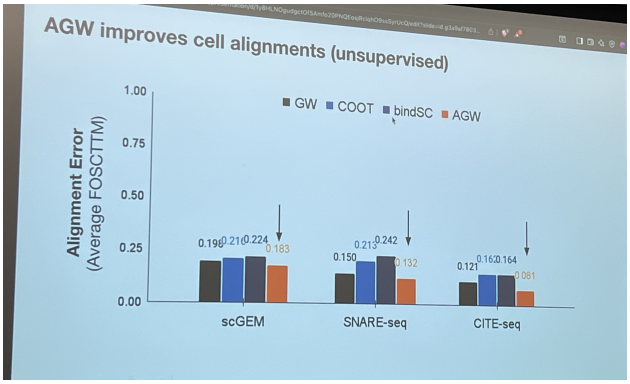


Figure 4: AGW improves cell alignments (unsupervised)

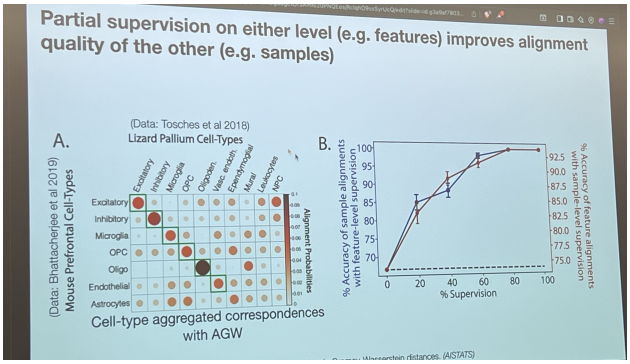


Figure 5: Partial supervision on either level (features or cells) improves alignment quality of the other (e.g. samples).

AGW align cells types for clusters more often than GW, especially in the unbalanced case.

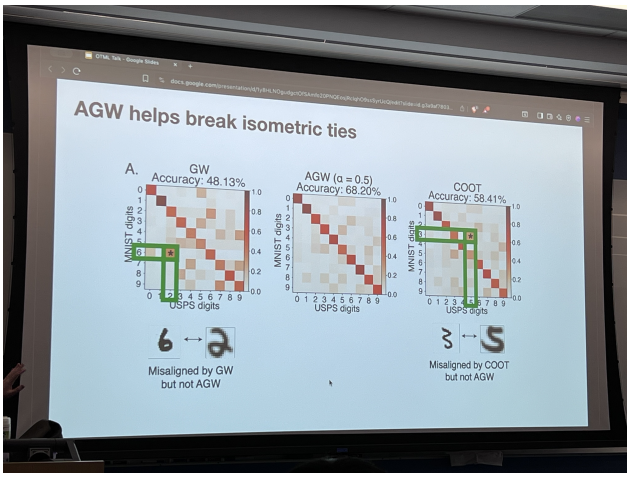


Figure 6: AGW help breaks isometric ties.

1.6 Fused GW

Spatial transcriptomics measures both gene expression and coarse spatial coordinates for tissue slices; fused GW is used to align these slices in a way that respects *both* spatial structure and expression patterns. (Zeira et al., 2022) 3D tissue block $\longrightarrow$ thin slices ($\sim 10\,\mu$m) placed onto a plate.

For each slice:

- 2D coordinates (array spot locations),
- Gene expression (transcript counts).

Key points:

- Tissue is thinly sliced into multiple slices and placed onto a plate.
- Plate has spots that measure gene expression of cells right on top, and also record spatial coordinates via "DNA barcodes".
- **Coordinates do not correspond to the location in the tissue.**

By aligning all slices with fused GW, we reconstruct a coherent 3D representation of the tissue that can be used for spatial DE, clustering, feature generation and cell–cell interaction analysis.

Why?

- Discovering functional substructures.
- Spatial differential expression within cell types.
- Cell–cell interaction, microenvironment effects.

However, slices are never align perfectly due to 3-D structures and human errors in slicing. Therefore, we prefer using partial alignment. For each consecutive pair of slices, fused GW minimizes a combination of gene-expression dissimilarity and spatial dissimilarity, under a *partial* OT constraint that only aligns a fraction s of spots.
For each consecutive pair of slices, solve

$$\min_{\Gamma \in \Pi(\mu,\mu',s)} (1-\alpha) \sum_{i,j} c(x_i, x'_j)\, \Gamma_{ij} \;+\; \alpha \sum_{i,j,k,l} \left(D_{ik} - D'_{jl}\right)^2 \Gamma_{ij}\, \Gamma_{kl},$$

gene expression dissimilarity spatial dissimilarity

where the partial OT constraint set is

$$\Pi(\mu, \mu', s) = \left\{ \Gamma \in \mathbb{R}_+^{n \times n'} : \Gamma \mathbf{1}_{n'} \leq \mu,\ \Gamma^\top \mathbf{1}_n \leq \mu',\ \sum_{i,j} \Gamma_{ij} = s \right\},$$

so that we *align a fraction s of spots*.

Zeira et al. (2022) Alignment and integration of spatial transcriptomics data. Nature Methods.
Liu et al. (2023) PASTE2: Partial Alignment of Multi-slice Spatially Resolved Transcriptomics Data. Genome Research.

1.7 Understanding the drivers of dynamics underlying cell populations

Recall OT is used in biology in the following aspects:

1. **Aligning cells** across modalities, time points or tissue slices to overcome experimental limitations. Here, the main object is a *transport map* or coupling, which tells us how to match individual cells or spots.
2. **Comparing distributions** across species or conditions using the OT distance, which is natural because biological data have a meaningful metric structure (e.g. in gene-expression space or latent space).
3. **Defining a consensus / reference map** over samples from different individuals or different slices of the same tissue via OT *barycenters*, which aggregate multiple distributions into a single representative one.

We will focus on part 2: Learning continuous-time dynamics

- evolving cell populations
- tissue architecture changes from snapshots

The first major application is in understanding the drivers of dynamics underlying cell populations.
Many biological systems are characterized by cells transitioning between functional states:

- Cellular differentiation in development
- Pluripotent reprogramming of cells
- Disease progression / drug resistance

The underlying dynamical process is encoded in the gene regulatory interactions.

- In 1950s, biologist Conrad Waddington compared cell differentiation to *"marbles rolling down a hill"* on a landscape ("Waddington's landscape").
- Landscape is structured by gene regulatory states.
- Can pull on the "wires" (regulatory interactions) to reshape it.

We have the following **Questions of interest:**

1. Can we predict which early cell states will give rise to which cell fates?
2. What are the gene regulatory mechanisms that drive different trajectories?
3. How can we intervene to re-route trajectories into desired targets?
4. How does the spatial context influence cell trajectories?

However, we also have **Challenges:**

- Single-cell sequencing is destructive:

  - **Cannot track a cell across time.**
  - All temporal correspondences are lost.
  - Observe a subset of the population at any given time.

- Cells measured at the same time point usually exhibit **asynchronous transitions**.
- Sampling can be **irregular in time**.

Since cells are growing across time, we need methods to adapt such changes, including time-point matching, consecutive discrete maps. Formally, we have the following families of methods.

Families of approaches:

1. Discrete maps between consecutive snapshots
   (a) Various OT approaches
2. Gradient flows: parameterizing the underlying landscape
3. Parameterizing the vector field:
   (a) Dynamical OT + neural ODEs
   (b) Flow matching etc

1.7.1 GeneTrajectory: "transporting" genes

Specifically, we look at one approach– **GeneTrajectory**.

GeneTrajectory builds a gene–gene distance using OT over the *cell* graph (treating each gene as a distribution over cells), then embeds genes in a low-dimensional space to reveal gene trajectories. The idea of GeneTrajectory is to study how genes themselves evolve along cellular trajectories.
Gene–gene OT distances (over the cell graph).

Define the ground cost on cells

$$C(i, j) = d_G(i, j),$$

where d_G is a distance on the cell graph.

For each gene g_a, define a distribution over cells

$$\rho_a(i) = \frac{g_a(i)}{\sum_{j=1}^{m} g_a(j)},$$

i.e. normalized expression of gene g_a across the m cells.

The OT distance between two genes g_a and g_b is

$$W(\rho_a, \rho_b) = \min_{\Gamma \in \Pi(\rho_a, \rho_b)} \sum_{i,j} \Gamma(i, j)\, C(i, j),$$

where $\Pi(\rho_a, \rho_b)$ is the set of couplings with marginals ρ_a and ρ_b.

Computing $W(\rho_a, \rho_b)$ for many gene pairs yields a gene–gene OT distance matrix, which can be embedded into low dimensions (e.g. via diffusion maps) to visualize gene trajectories.

Qu et al. (2025) "Gene trajectory inference for single-cell data by optimal transport." Nature Biotechnology.

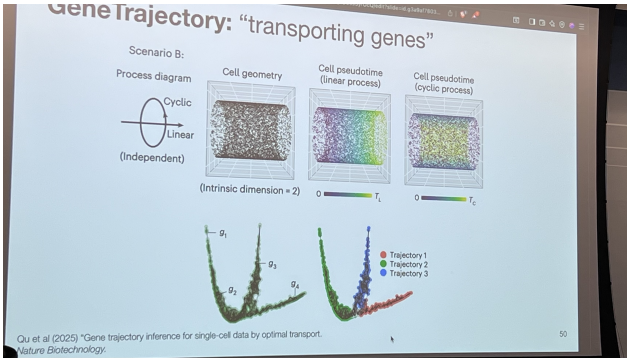


Figure 7: Linear: differentiate cell types/dividing. Cyclic: cell cycle/growing.

1.7.2 Gradient-Flow and Dynamical OT Models

Modeling the landscape with gradient flows

We treat cell distributions as moving along the steepest descent of an energy functional $J(\rho)$ in Wasserstein space; each JKO step updates ρ_t by minimizing J plus a W_2^2 penalty, and we then fit J so that the resulting trajectory matches the observed snapshots.

JKO step:

$$\rho_{t+1} = \arg\min_{\rho \in \mathcal{P}_2(X)} J_\varepsilon(\rho) + \frac{1}{2\tau} W_2^2(\rho_t, \rho).$$

At each step we compute a loss comparing the predicted next distribution to the observed data:

$$\text{loss} = W_\varepsilon\big(\text{data}_{t+1}, \widehat{\rho}^{\varepsilon}_{t+1}\big).$$

$\Rightarrow$ Taking a step in the direction of the steepest gradient to get the next time point.

Caveat: gradient flows do not account for cell cycle.

Modeling the landscape with FGW gradient flows

FGW gradient flows extend the previous approach by using fused Gromov–Wasserstein, so that the learned landscape accounts for both gene-expression features and spatial information.

Stories:

- Use **FGW** to take into account the spatial information.
- Parameterize the energy J with an **MLP** instead of ICNNs.

Bunne et al. (2022) Proximal Optimal Transport Modeling of Population Dynamics. AISTATS.

Learning the flow of a distribution via a velocity field

Instead of updating distributions via JKO steps, we can describe their evolution by a time-dependent velocity field $v_t(x)$. Single cells follow $\dot{x}_t = v_t(x_t)$ and the population density obeys the continuity equation with boundary conditions ρ_0, ρ_T, a flow-matching type formulation.

Initial and final distributions in $\mathbb{R}^d$:

$$X_0 \sim \rho_0 \in \mathbb{R}^d, \qquad X_T \sim \rho_T \in \mathbb{R}^d.$$

Time-dependent velocity field $v_t(x)$.

- For a single particle / cell x:

$$\frac{dx_t}{dt} = v_t(x_t).$$

- For distributions over cells:

$$\frac{\partial \rho_t(x)}{\partial t} = -\nabla \cdot \big(v_t(x)\, \rho_t(x)\big).$$

With boundary conditions:

$$\rho_t(x)\big|_{t=0} = \rho_0(x), \qquad \rho_t(x)\big|_{t=T} = \rho_T(x),$$

and normalization

$$\int_{\mathbb{R}^d} \rho_0(x)\, dx = \int_{\mathbb{R}^d} \rho_T(x)\, dx = 1.$$

Challenge: Continuity equation has non-unique solutions.

Restricting velocity fields by learning the velocity field with dynamical OT

Define conditional path ahead of time, dynamical OT is taking the most efficient path between the samples (conditional flow matching)

Reconstructing spatial transcriptomic snapshots. Errors accumulate across time. We model the whole time point together to define the path.
Selecting a velocity field such that the overall path taken is minimized:

$$\mathcal{L}_W = \min_{v(x,t)} \int_0^T \int_{\mathbb{R}^d} \big\|v_{t,\theta}(x)\big\|^2\, \rho_t(x)\, dx\, dt \quad \text{(minimizing "kinetic energy")},$$

where $v_{t,\theta}$ is parameterized by a neural network.

This can be written as

$$\mathcal{L}_W = T\, \mathbb{E}_{z_0 \sim \rho_0} \int_0^T \big\|v_{t,\theta}(x_t)\big\|^2 dt.$$

Subject to the same continuity equation and boundary conditions as before:

$$\partial_t \rho_t(x) + \nabla \cdot \big(v_t(x) \rho_t(x)\big) = 0,$$

$$\rho_t(x)\big|_{t=0} = \rho_0(x) \quad \text{and} \quad \rho_t(x)\big|_{t=T} = \rho_T(x).$$

Sha et al. (2024) "Reconstructing growth and dynamic trajectories from single-cell transcriptomics data." Nature Machine Intelligence.